

HIMANSHU BADHANI

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🌐 [website](#)

RESEARCH INTERESTS

My primary interests are open quantum systems, with a focus on understanding thermodynamic and information-theoretic properties of noisy processes.

Topics I have explored or am currently studying:

- Non-Hermitian quantum mechanics, with a focus on the formalism of pseudo-Hermitian quantum mechanics.
- Quantum information theory, with a focus on information-theoretic properties of the quantum channels.
- Quantum thermodynamics, with a focus on thermodynamic resource theory of quantum processes.
- Algebraic quantum mechanics, the structure of C^* -algebras.
- Geometric features in quantum mechanics, specifically formalizing thermodynamics in a “curved” Hilbert space.

Topics that I plan to explore in near future:

- Understanding the dynamical causal structure in many body systems.
- Quantum simulations using open quantum system techniques.

EDUCATION

Ph.D.

2018-2025

The Institute of Mathematical Sciences, Chennai, India.

Thesis: *Subsystem dynamics in pseudo-Hermitian quantum mechanics*.

Submitted on November 27, 2024.

Defended on August 04, 2025.

Advisor: Dr. C M Chandrashekar.

BS-MS

2011 - 2016

Indian Institute of Science Education and Research (IISER), Pune, India.

Thesis: *Quantum fields in de-Sitter space*

CGPA: 8.2/10

Advisor: Dr. Alok Laddha, CMI Chennai

EMPLOYMENT

Visvesvaraya postdoctoral fellow at the Center for Quantum Science and Technology,
IIIT Hyderabad

Feb 2025–Present

Research internship

2016–2017

Harish-Chandra Research Institute, Allahabad, India.

Project title: *de-Sitter stability and the inflationary potential*

Advisor: Dr. Dileep Jatkar

Gap year: In 2017-2018 I took a gap year and explored small scale farming in my ancestral village. I am passionate about reforestation efforts in Himalayas and I am experimenting with low intervention farming. I also got interested in the mathematical theory of finance and briefly joined the doctoral program in finance at IIM Kolkata, a premier management institute in India, in 2018.

PUBLICATIONS

[Google scholar](#) / [ArXiv](#)

1. **HB** and S. Das, Conditional channel entropy sets fundamental limits on thermodynamic quantum information processing, arXiv:2604.01217 [quant-ph] (2026)
2. **HB**, Dhanuja G. S, and S. Das, Thermodynamic work capacity of quantum information processing, arXiv:2510.23731 [quant-ph] (2025).

3. **HB**, Dhanuja G. S, and S. Das, Thermodynamics of quantum processes: An operational framework for free energy and reversible athermality, arXiv:2510.12790 [quant-ph] (2025).
4. **HB**, Dhanuja G. S., S. Choudhary, V. Anand, and S. Das. Erasure cost of a quantum process: a thermodynamic meaning of the dynamical min-entropy, *Quantum Sci. Technol.* **11** 015038, (2026).
5. **HB**, C. M. Chandrashekar. Reduced dynamics in quasi-Hermitian systems, *Annals of Physics, Volume 484*, 170304, (2026).
6. R. Basu, A. Chakraborty, **HB**, M. Alimuddin, and S. Bhattacharya. Exploring non-markovianity in ergodic channels: Measuring memory retention through ergotropy. *Phys. Rev. A* **111**, 032416, (2025).
7. F. Buscemi, R. Gangwar, K. Goswami, **HB**, T. Pandit, B. Mohan, S. Das, and M. N. Bera. Causal and noncausal revivals of information: A new regime of non-markovianity in quantum stochastic processes. *PRX Quantum* **6**, 020316, (2025).
8. **HB** and S. Ghosh. Decomposition of a system in pseudo-hermitian quantum mechanics. *J. Phys. A: Math. Theor.* **58** 165301, (2025).
9. **HB**, S. Banerjee, and C. M. Chandrashekar. Non-Hermitian quantum walks and non-Markovianity: the coin-position interaction. *Phys. Scr.* **99** 105112, (2024).
10. A. Mitra, **HB**, and S. Ghosh. Improvement in quantum communication using quantum switch. *Phys. Scr.* **98** 045101, (2023).
11. **HB** and C. M. Chandrashekar. Gravitationally induced entanglement dynamics between two quantum walkers. *Eur. Phys. J. C* **81**, 454, (2021).

CONTRIBUTED TALKS AND SEMINARS

1. (Upcoming) Physics seminar at IMSc Chennai, May 2026, **Title:** Thermodynamic aspects of quantum processes.
2. [Physics seminar](#) at TIFR Hyderabad, May 2026, **Title:** Thermodynamic aspects of quantum processes.
3. Physics seminar at IIIT Hyderabad, Jan. 2024, **Title:** Virtual subsystems of non-trivial inner product spaces.
4. [Chennai Strings Meeting 2023](#), IMSc, Chennai, **Title:** Partial tracing in a non-trivial metric Hilbert space.
5. [PHHQP20 2023](#), Koc University, Istanbul, **Title:** PT-symmetric quantum walk: exploring the reduced dynamics.
6. [PiQUST 2023](#) IIT Madras, **Title:** Non-Hermitian quantum walks.
7. [IMSc60 2023](#), IMSc Chennai, **Title:** Non-Hermitian quantum walks.

NUMERICAL SKILLS

- Languages: MATLAB, Mathematica, and Python.
- Numerical skills: Monet Carlo optimization, classical simulated annealing, semidefinite programming.
- QI tools: Qiskit, CVXPY, PennyLane

TEACHING EXPERIENCE

- Tutoring:
 - * Quantum Information Theory by Dr. Siddhartha Das, Monsoon 2025
 - * Introduction to Quantum Information by Dr. C. M. Chandrashekar, Spring 2020
- I have also mentored a few undergraduate summer students at IMSc over the years. I have been mentoring with a few young PhD students at IIIT Hyderabad and along with this I occasionally take classes/ prepare question papers for undergraduate courses.

PEER REVIEW AND OUTREACH

- **Peer review:** Class. Quantum Grav. (1), J. Phys A (1), Physica Scripta (2), QIP (1), TQC 2024 (1).
- Panel member in the workshop on [Impact of Quantum Technologies on Life Sciences](#), hosted by IIIT Hyderabad and organized with NITI Aayog, Gov. of India.

- YouTube manager for [Quantum Meets](#).
- Among the early members of [Disha](#) IISER Pune, a volunteer organization working on educational equity. I also started and ran a series of talks called *Talk for Twenty* and a magazine called *Pahal*, both of which are still going strong at IISER Pune.

ACADEMIC MERITS AND AWARDS

- [Visvesvaraya](#) Post-Doctoral Fellowship (2025-2026) by the Ministry of Electronics and Information Technology (MeitY), Gov. of India.
- Achieved the national rank 22 in the **Joint Entrance Screening Test (JEST) 2018** for PhD positions in India.
- Qualified for the Junior Research Fellowship by CSIR, Gov. of India.
- INSPIRE fellowship (2011-2016) by the Dept. Of Science and Technology, Gov. of India

REFERENCES

Dr. C M Chandrashekar [✉](#)
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 Bangalore
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 Institute of Mathematical Sciences,
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 Technology, Hyderabad
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